Air Fryer Market Demand Estimation Report

Jun Hong & Derek Wang

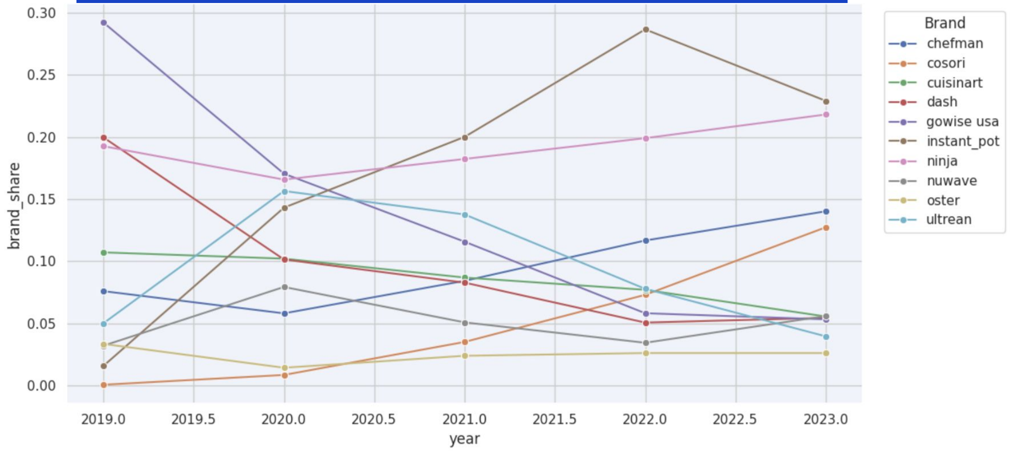
The Airfryer Market at a Glance

10
Brands tracked
2019-2023

Ninja
Market Leader
19.2% avg. Share

$57-224
Price Range
Avg. Price by brand

Average Market Share by Brand



Key Market Dynamics

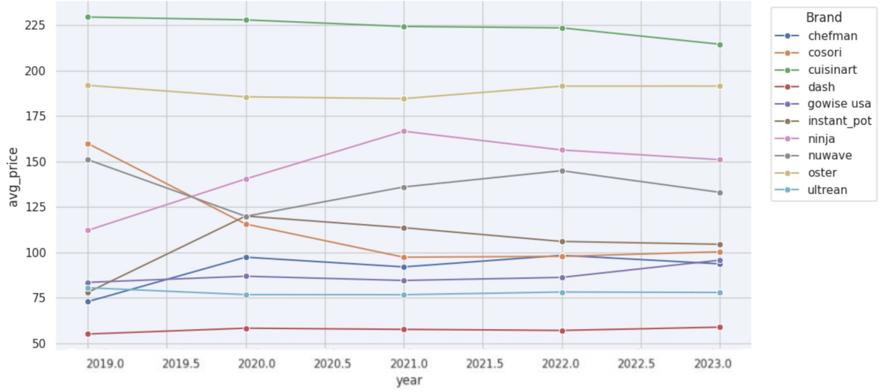
Ninja and Instant Pot together command ~37% of the market, making them the clear volume leaders.

GoWise USA and Dash were early share leaders but lost ground sharply after 2020 as Ninja accelerated.

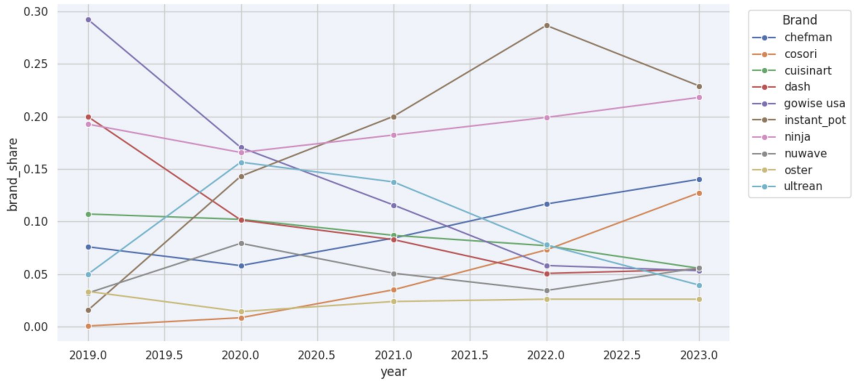
Cuisinart and Oster occupy the premium tier ($190–$224) but hold modest shares, suggesting limited mass-market appeal at high price points.

Price, Ratings & Market Trends

Average Price by Brand



Market Share by Brand (%)



Average Customer Rating by Brand (stars)

How we model Demand

Model: Multinomial Logit Demand (estimated by OLS on log market shares)

$$\log(s_{bt}) = \alpha_0 + \alpha_t + \gamma_b + \beta_{price}\, p_{bt} + \beta_{rating}\, r_{bt} + \sum_{\ell=1}^{L} \beta_\ell x_{bt\ell} + \varepsilon_{bt}$$

Why Log Shares?

The logit model predicts that log market share is linear in price and product attributes. This lets us estimate demand with standard linear regression

Year Fixed Effects

Year dummies absorb any market-wide shocks (e.g., pandemic cooking boom in 2020). They isolate brand and price effects from economy-wide trends.

Brand fixed effects

Brand dummies capture persistent quality
and reputation differences not explained by observed features. They prevent brand loyalty from biasing the price coefficient.

Demand Drivers: Price Sensitivity & Product Features

Price coefficient

β_price = -0.0377

A $10 price increase
reduces log share by 0.38

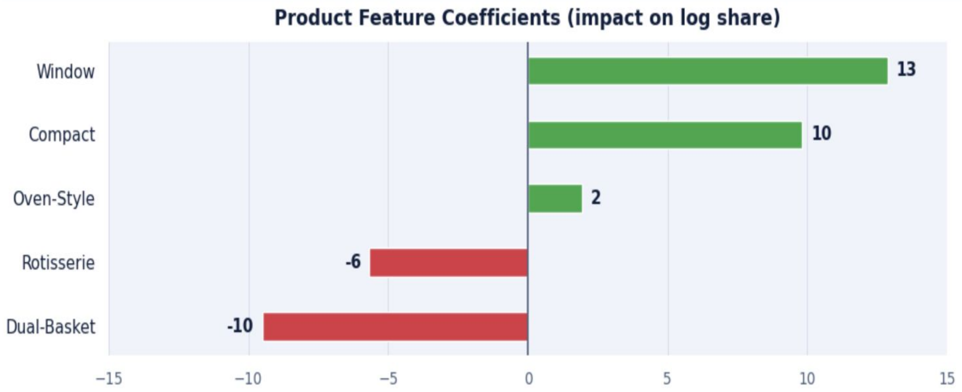
Why the negative sign matters

The price coefficient must be negative for the demand model to be logically consistent.

✔ Economic validity: higher prices reduce quantity demanded

✔ Markup formula: a negative β gives a positive markup — essential for cost recovery

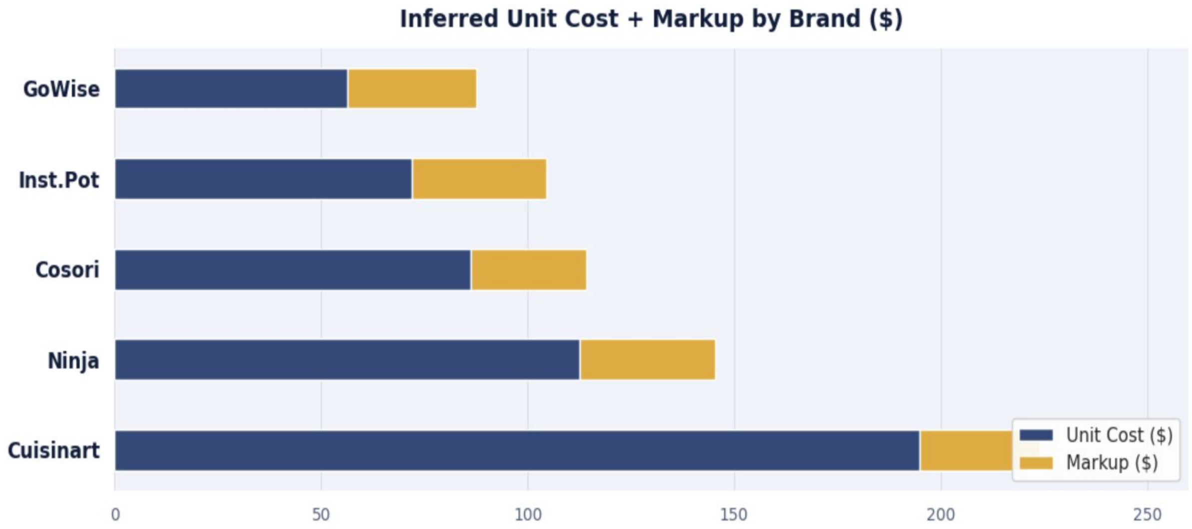
✔ Profit analysis: only with correct sign can we infer meaningful unit costs and markups

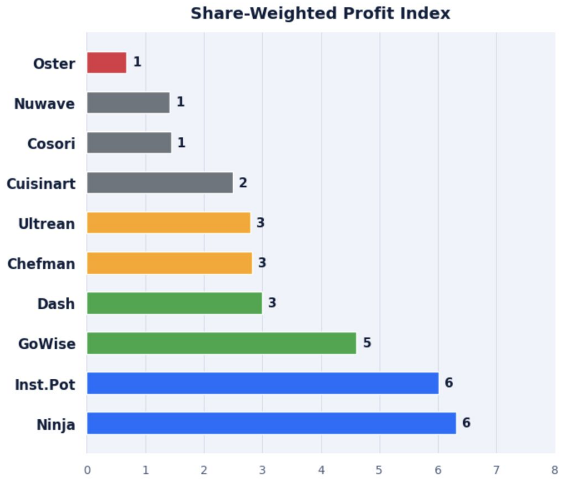


Window & compact
features strongly boost share — consumers prefer visibility and space efficiency.

Dual-basket & rotisserie
reduce share — niche features that reduce perceived simplicity.

Unit Costs, Markups & Share-Weighted Profit





Brand	Avg. Price	Unit Cost	Markup	Profit Index
Ninja	$145	$113	$32.9	**6.31**
Instant Pot	$105	$72	$32.6	**6.01**
GoWise USA	$88	$56	$31.2	**4.61**
Dash	$58	$28	$29.5	2.99
Chefman	$91	$62	$29.4	2.82
Ultrean	$78	$49	$29.3	2.80
Cuisinart	$224	$195	$29.1	2.50
Cosori	$114	$86	$28.0	1.44
Nuwave	$137	$109	$28.0	1.42
Oster	$189	$162	$27.2	0.68

Brand Competitive Positions

Volume Leaders

Ninja (19.2%) · Instant Pot (17.5%) · GoWise USA (13.8%)

High share, mid-to-high price, top profit indices. Ninja is the clear category king with growing share and premium pricing. Instant Pot leverages brand trust from pressure cookers.

Value Players

Dash ($57) · GoWise USA ($87) · Chefman ($91)

Affordable entry-level brands with strong share among price-sensitive buyers. Despite low prices, markups are comparable to rivals ($28–$31) thanks to low cost bases.

Premium Niche

Cuisinart ($224) · Oster ($189) · Nuwave ($137)

High list prices and high unit costs, but modest market share. Premium positioning does not translate into outsized profit when volume is limited.

Rising Challengers

Cosori (4.66★) · Ultrean (4.67★) · Chefman

Brands with above-average ratings but below-average share. If they convert satisfaction into volume, they could move up in profit rankings.

Conclusions and Strategic Takeaways

1

Ninja dominates and is pulling away

With 28% share in 2023 (up from 10% in 2019) and the highest profit index, Ninja has established category leadership. Its premium pricing is supported by strong brand equity, not just features.

2

Price sensitivity is real but moderate

A $10 increase in price reduces log share by ~0.38 points. Brands can charge more if they build perceived quality, but value-seekers remain a large segment (Dash/GoWise).

3

Simplicity sells — window & compact win

Consumers reward visibility (window) and compactness. Complex features like dual-basket and rotisserie carry demand penalties, suggesting most buyers want a simple, reliable appliance.

4

Markups are surprisingly uniform

All brands earn $27–$33 per unit. The real profit gap comes from volume. High-share brands generate outsized profit because the margin structure is similar across the board.

5

Premium brands face a scale problem

Cuisinart and Oster charge 2–3× more, but their market shares (~3–9%) limit total profit. Without a path to higher volume, premium positioning is a strategic trap in this category